

MOHAMMAD ASHRAFUL ISLAM

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ACADEMIC CREDENTIALS

- **Bachelor of Science in Mechanical Engineering**
- Chittagong University of Engineering & Technology (CUET), Chittagong, Bangladesh
- March 2020 - July 2025
- CGPA: 3.45/4.00 (3.84/4.00 in final year)

RESEARCH INTERESTS

Computational Fluid Dynamics (CFD) | Materials & Metallurgy | Heat Transfer | Combustion | IC Engine | Multiphase Granular Flow | Solid-Liquid Interaction

PUBLICATIONS

Journal Articles

1. M. A. Islam, Kazol, M. H., Rahman, M., Al Rifat, A., & Amin, N. (2026). A comprehensive review of recent advances in biomass pyrolysis: feedstock characteristics, thermal decomposition mechanism, temperature, heating rate, residence time, and particle size on product distribution. *Future Energy*, <https://doi.org/10.55670/fpll.fuen.5.2.5>.
2. M. A. Islam, Kazol, M. H., Rahman, M., Rahman, A., & Meraz, H. M. (2026). Numerical investigation of biomass thermochemical conversion in a fluidized bed reactor: effect of temperature and particle size on product distribution using Eulerian-Lagrangian Multiphase with DPM injection and KTGF. (Under Review).
3. Haque, M. N., Al Rifat, A., Hassan, T., Alam, M. I., Islam, M. A. Islam, Kazol, M. H., Adib, A. R., & Rahman, M. (2026). Investigation of the effects of piston bowl geometry and spray angle on combustion, performance, and emission of a diesel engine. *Energy Science & Engineering* (Accepted in *Energy Science & Engineering Journal*).
4. Kazol, M. A. Islam, M. A., Rahman, M., & Rahman, M. A. (2026). A computational investigation of varying spray angles on a dual-fuel engine with modified re-entrant piston bowl geometry. *Heliyon* (Under Review).
5. A. Bakkar, M. A. Islam, S. C. Banik. (2026). Experimental Investigation of Wood-Based Piezoelectric Materials for Sustainable Energy Harvesting. (Under Review in *Next Materials*)

Conference Proceedings

5. M. A. Islam, Rahman, M., & Bakkar, A. (2025). Fabrication and performance analysis of automobile brake pads, using eco-friendly and recycled materials. *SciEn Conference Series* (Accepted, in production).

RESEARCH & PROFESSIONAL EXPERIENCE

Research Assistant | Bio-Fuel & Energy Lab, CUET

Sep 2025 - Present

- Principal Investigator: Prof. Dr. Md. Mizanur Rahman, Department of ME
- Numerical and experimental investigation of gas-solid reactive flows in high-temperature thermochemical conversion processes.
- Developed and implemented ANSYS Fluent, Converged CFD, and OpenFOAM-based multiphase CFD models to simulate reactive gas-solid interactions, species transport, and conjugate heat transfer in fluidized bed reactors.
- Investigated the coupled effects of temperature, particle size, and residence time on gas-phase species distribution and solid conversion rates.
- Validated numerical predictions against experimental data, establishing a framework for physics-based reduced-order modeling of diffusion-reaction mechanisms.

Undergraduate Thesis Researcher | CUET

Topic: Fabrication and performance analysis of automobile brake pads using eco-friendly and recycled materials

- Designed and tested high-temperature materials subjected to thermal and mechanical loading.
- Analyzed material degradation mechanisms under frictional heating and cyclic thermal stress.

Intern | Super Petrochemical Limited (SPL), Chittagong, Bangladesh

Duration: 3 weeks

- Naphtha processing unit - high-temperature gas-solid handling and thermal systems.
- Observed and assisted in naphtha reforming furnaces operating at elevated temperatures, gaining exposure to industrial-scale gas-solid thermal interactions and heat transfer management.
- Participated in rotating equipment maintenance (compressors, turbines) and static equipment inspection (pressure vessels, heat exchangers) with emphasis on high-temperature material performance.
- Documented fabrication and welding procedures for petrochemical components, understanding thermal stress effects on metallic structures.
- Reviewed process control systems for monitoring gas-phase species and temperature profiles in reforming reactors.

TECHNICAL SKILLS

- OpenFOAM: Multiphase solvers, custom boundary conditions, species transport, conjugate heat transfer
- CAD & Simulation: ANSYS, ANSYS Fluent, ANSYS Forte, SOLIDWORKS, AutoCAD, Aspen Plus, Converged CFD
- CFD Post-processing: Tecplot, ParaView, OriginPro
- Programming: Python, MATLAB, C, Fortran
- Basic Tools: LaTeX, Adobe Illustrator, Adobe Photoshop

ACHIEVEMENTS

- Merit scholarship, Department of Mechanical Engineering, CUET
- Scholarship (7th place in region), Junior School Certificate (JSC)

REFERENCES

Dr. Md. Mizanur Rahman

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